

Linux AXI Ethernet driver

Xilinx Wiki

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Table of Contents

1	Table of Contents	5
2	HW IP features	6
2.1	AXI 1G/2.5G Ethernet Subsystem (PG138)	6
2.2	10 Gigabit Ethernet subsystem (PG157)	6
2.3	10G/25G Ethernet Subsystem (PG210)	6
2.4	USXGMII Ethernet Subsystem (PG251)	7
2.5	MRMAC Ethernet Subsystem (PG314)	7
2.6	Switchable 1/10/25G IP support (PG292)	8
2.7	DCMAC ethernet subsystem (PG369)	8
3	Features supported in the driver	9
4	Missing Features and Known Issues/Limitations in Driver	10
4.1	Missing Features and Known Issues/Limitations in SDT	11
4.1.1	Important AR links	15
5	Kernel Configuration	16
6	Device-tree	17
6.1	Soft Ethernet MAC Configured with MCDMA	17
6.1.1	(1G, legacy 10G or 10G/25G MAC, MRMAC) with MCDMA	17
6.1.2	DCMAC with MCDMA	19
6.2	Related device tree information	22
7	IEEE 1588 Support	23
7.1	SWITCHABLE 1/10/25G IP	24
8	Performance	27
8.1	1G Ethernet with AXIDMA	27
8.2	10G Ethernet with AXIMCDMA	28
8.3	Diagnostic and Protocol Tests	29
8.4	Stress Test	30
8.5	Performance Test	30

8.6	1588 Testing	31
9	Mainline status.....	33
10	Change Log	34
11	Related Links	42
12	Linux Ethernet Offload Engine driver	43
12.1	Table of Contents	43
12.2	Introduction	43
12.3	HW IP features	43
12.3.1	Features supported in the driver	43
12.4	Missing Features and Known Issues/Limitations in Driver.....	44
12.4.1	Missing Features and Known Issues/Limitations in SDT	44
12.5	Kernel Configuration.....	44
12.6	Device-tree.....	45
12.6.1	EOE Configured with MCDMA	45
12.7	Performance.....	47
12.7.1	10G Ethernet with AXIMCDMA.....	47
12.7.2	Performance Test	47
12.8	Mainline status.....	48
12.9	Change Log	48
12.10	Related Links	48

Axi Ethernet Linux driver for Microblaze, Zynq, Zynq Ultrascale+ MPSoC and Versal

Introduction

1 Table of Contents

This page gives an overview of Axi Ethernet Linux driver which is available as part of the Linux distribution. Paths, files, links and documentation on this page are given relative to the Linux kernel source tree.

2 HW IP features

2.1 AXI 1G/2.5G Ethernet Subsystem (PG138)

- Support for MII, GMII, RGMII, 1G/2.5G SGMII, and 1000/2500 BASE-X PHY interfaces
- Support for 2.5G Ethernet. In 2.5G mode, only SGMII and 2500BASE-X PHY interfaces are available.
- Support for 1000BASE-X and SGMII over select Input/Output (I/O) low voltage differential signaling (LVDS)
- Support for pause frames for flow control
- Media Independent Interface Management
- Ethernet Audio Video Bridging (AVB) support
- AXI4-Stream transmit/receive interface
- IEEE Standard 1588 Support
- AXI4-Lite register interface

2.2 10 Gigabit Ethernet subsystem (PG157)

- Designed to 10 Gigabit Ethernet specification IEEE Standard 802.3-2012
- AXI4-Stream protocol support on client TX and RX interfaces
- Configured and monitored through an optional AXI4-Lite Management Data interface or using status and configuration vectors
- Supports 10GBASE-SR, -LR and -ER optical links in Zynq-7000, UltraScale™, Virtex-7, and Kintex-7 devices (LAN mode only)
- Supports 10GBASE-KR backplane links including Auto-Negotiation (AN), Training and Forward Error Correction (FEC)
- Supports Deficit Idle Count
- Comprehensive statistics gathering
- Supports 802.3 and 802.1Qbb flow control
- Supports VLAN and jumbo frames
- Custom Preamble mode
- Independent TX and RX Maximum Transmission Unit (MTU) frame length
- Supports high accuracy IEEE Standard 1588-2008 1-step and 2-step timestamping on a 10GBASE-R network interface

2.3 10G/25G Ethernet Subsystem (PG210)

- Designed to the Ethernet requirements for 10/25 Gb/s operation specified by IEEE 802.3 Clause 49, IEEE 802.3by, and the 25G Ethernet Consortium

- Includes complete Ethernet MAC and PCS/PMA functions or standalone PCS/PMA for 25Gb/s operation
- Includes complete Ethernet MAC and PCS/PMA functions, standalone MAC or standalone PCS/PMA for 10 Gb/s operation
- Simple packet-oriented user interface
- Comprehensive statistics gathering
- Status signals for all major functional indicators
- BASE-R PCS sublayer operating at 10.3125Gb/s or 25.78125Gb/s
- Optional clause 74 BASE-KR FEC sublayer
- Optional Auto-Negotiation
- Optional clause 108 25G Reed-Solomon Forward Error Correction (RS-FEC) sublayer
- Custom Preamble mode
- Optional IEEE 1588 1-step and 2-step timestamping
- Runtime switchable between 10G and 25G

2.4 USXGMII Ethernet Subsystem (PG251)

- Designed to meet the USXGMII specification EDCS-1467841 revision 1.4
- Supports 10M, 100M, 1G, 2.5G, 5G, or 10GE data rates over a 10.3125 Gb/s link
- Both media access control (MAC) and PCS/PMA functions are included
- Code replication/removal of lower rates onto the 10GE link
- Rate adaption onto user clock domain
- Low data path latency
- 32-bit AXI4-Stream interface for datapath
- Optional AXI4-Lite register interface
- Support for 802.3x and priority-based pause operation
- Detailed statistics gathering
- Support for custom preambles
- Supports deficit idle count (DIC)

2.5 MRMAC Ethernet Subsystem (PG314)

- Hardened Ethernet IP block on Versal.
- Multi rate Ethernet MAC supporting speeds from 10G to 100G.
 - The driver supports 25GE and 10GE with 1 to 4 lanes.
- Hardened IP (to be used with Soft DMA and logic for driver subsystems)
- High performance, low latency.
- Low data path latency
- User-side AXI4-Stream interface for data
- AXI4-Lite register interface
- Detailed statistics gathering
- IEEE1588 support

2.6 Switchable 1/10/25G IP support (PG292)

- Designed to the Ethernet requirements for 1/10 Gb/s operation specified by IEEE 802.3 Clause 49 or Clause 36
- Runtime switchable Ethernet MAC and PCS/ PMA functions for 1/10/25 Gb/s operation.
- Supports only GTHE3/GTYE3 and GTHE4/GTYE4 transceiver supported devices
- Include Auto Negotiation (Clause-73)
- Enable Link Training (Clause 72)

2.7 DCMAC ethernet subsystem (PG369)

- Hardened Ethernet IP block on Versal Premium and Versal HBM (to be used with Soft DMA and logic for driver subsystems).
- 1 x 400GE, 3 x 200GE, 6 x 100GE, or combinations of 100 Gbps, 200 Gbps, and 400 Gbps totaling up to 600 Gbps support.
 - Driver supports 100G with single lane (1 x 100GE).
- Supports 100GBASE-R, 200GBASE-R and 400GBASE-R phy interfaces.
- 40-channel time-sliced MAC capable of 600 Gbps operation.
 - Channelized option for time-sliced applications.
 - Up to 40 channels supported.
 - User-defined bandwidth allocation granularity.
- Pause frame processing, including priority-based flow control.
- Optional built-in RS-FEC functionality.
- IEEE1588 support.
- User-side segmented AXI4-Stream interface.
- AXI4-Lite register interface.
- Detailed statistics gathering.

3 Features supported in the driver

- Support following ethernet IPs:
 - AXI 1G/2.5G Ethernet subsystem (PG138)
 - 10G Ethernet subsystem(PG157)
 - 10G/25G Ethernet Subsystem(PG210)
 - USXGMII(PG251)
 - MRMAC(PG314)
 - Switchable 1/10/25G(PG292)
 - DCMAC(PG369)
- IEEE 1588 Support for 1G and legacy 10G MAC (PG157), 10G Ethernet subsystem and 25G Ethernet subsystem (PG210) and MRMAC
- Speed support for 10/100/1000 Mbps for 1G MAC
- 10G Base-R support for Legacy 10G MAC(PG157) and 10G MAC (PG210)
- 10G and 25G speed support for MRMAC
- 100G with single lane support for DCMAC.
- Support for GMII/RGMII/SGMII/1000Base-X/2500Base-X/10GBase-R/25GBase-R/100GBase-R Phy Configurations
- Supports Independent 4K, 8K, 16K, or 32KB TX and RX frame buffer memory
- Support for common ethtool queries.
- NAPI support.
- Full/Partial Checksum offload support
- Support for Jumbo Frames
- Supports AXI DMA and AXI MCDMA dma configuration.
- Multi-queue support
- Switchable 1/10/25G IP supports 1G or 10G runtime switchable speeds in the driver.

4 Missing Features and Known Issues/Limitations in Driver

- The driver assumes that Axi Ethernet IP is connected to the DMA at the hardware level.
- The driver doesn't use dma engine framework and contains DMA programming sequence i.e doesn't use separate DMA driver. Hence compatibility string of axidma node (DTS) is set to a dummy device-tree property compatible = "xlnx,eth-dma";
- The driver doesn't support software time-stamping. It supports only hardware time-stamping.
- PTP synchronization along with high speed traffic (iperf or netperf) is not supported as under heavy load, timestamp in FIFO and DMA data in BD is expected to go out of sync and remain so until the interface is reset.
 - This is a limitation on Soft Ethernet designs which have HW timestamp FIFOs. This issue is not relevant to designs with inline HW timestamping.
- fixed-link mode not supported for AXI 1G/2.5G.
- For 1588 testing the Current driver assumes that AXI Stream FIFO is connected to the MAC TX Time stamp Stream interface at the design level. For axiethernet 1G/10G subsystem only 2-step PTP is supported.
- 10G/25G and USXGMII configurations do not support dynamic link status/change in the background as there is no external PHY using PHY framework.
- Pause frame solution is not supported and hence there could be RX overruns errors in bidirectional throughput.
- The driver supports MCDMA using kernel config i.e CONFIG_AXIENET_HAS_MCDMA option. So in multi-instance scenario driver will only support a single DMA type i.e 1G + MCDMA and 10G + MCDMA.
- The driver doesn't support extended multicast and VLAN support. Limited validation of multicast and vlan support.
- Runtime Switchable mode.
- MRMAC speeds 40G/50G/100G are not supported yet.
- MRMAC multi-lane support is not independent because if common GT reset logic exists in subsystem.
- AXI Ethernet is only validated on Zynq (1G), ZU+ (1G, 2.5G), Versal (1G, 2.5G) and MB (US+ and 7 series, 1G) platforms.
- USXGMII driver is only validated on ZU+ based platforms.
- XXV driver is only validated on ZU+, RFSoc and Versal based platforms (ZCU102, ZCU670, VCK190).
- MRMAC IP/driver is only available and validated on Versal based platforms (VCK190, VPK120)
- On versal (VCK190) support is limited to AXI 1G/2.5G Ethernet subsystem (without PTP), XXV Ethernet subsystem (without PTP, validated at 25G) and MRMAC.
- Switchable 1/10/25G IP support is only validated at 1G and 10G on Zynq Ultrascale+ MPSoC via ethtool (ZCU102).



NOTE: Relevant missing Features and Known Issues/Limitations in IP:

- Multiple TX and RX channel in MCDMA have common configuration and reset registers and hence cannot be used independently by multiple MACs. For ex., if XXV Ethernet instance 1 uses channel 0-4 of MCDMA and then XXV Ethernet instance 2 uses channels 5-15, then resets during driver initialization and error management effect all channels and both instance need to use common registers. Due to this limitation, multiple MACs cannot be used with a single MCDMA.
- Ethtool or link/speed information support is not available for MRMAC; this limitation is due to phylink framework and non-presence of an external PHY or PCS. However, there is a plan to add this support in an upcoming release.
- RSFEC configuration is not supported or tested in the driver.
- AXI Ethernet or any other high speed Ethernet does not have any module support with MCDMA. There are some dependencies.
- A kernel hang may be observed on some XXV designs when using 2024 releases. This is due to AN status register being inaccessible in the XXV IP and it will be fixed in a future release.
- phy-mode parameter is not automatically generated (by DTG or SDT) as 25000BaseX for 2.5G Ethernet designs as the IP does not provide this required information. This property needs to be edit manually in the DT. This should be fixed in a future release (2025)
- XXV Block lock register access in the IP fails (kernel crash with SError) when the reference GT clock is not stable (and link is not up). Since there is no indication to the SW on the clock stability, reads to this register cannot be avoided beforehand. If the clock is stable and link is up, there is no effect from this known issue. This should be fixed in a future release (2025)
- Supported and advertised link modes reported by ethtool for XXV are incorrect.
- DCMAC driver is validated on Versal premium platform.
- DCMAC driver supports only 1x100G configuration.
- IEEE 1588 support for DCMAC is not there. It will be added in future release.
- 2.5G SGMII phy mode support for AXI 1G/2.5G ethernet is not there.
- "max-speed" is required DT property for AXI 1G/2.5G as it is needed to distinguish 1G and 2.5G speed MAC types. Without this property, driver probe will fail.

4.1 Missing Features and Known Issues/Limitations in SDT

- MRMAC connected IP properties (such as GT control GPIOs) and Timer-Syncer IP properties (including associated timestamping related nodes, if any) are not automatically generated in SDT flow. This will be updated in a future release.
- Support to generate DCMAC and connected nodes (such as GT control and GT reset GPIOs) is not there and will be added in future release. Sample working DCMAC and connected nodes:

```

dcmac@a4000000 {
phy-mode = "100gbase-r";
xlnx,gt-mode-0 = "NRZ";
axistream-connected = <0x000000062>;
xlnx,num-queues = [00 01];
xlnx,channel-ids = "1";
interrupt-names = "mm2s_ch1_introut", "s2mm_ch1_introut";
interrupt-parent = <0x000000005>;

```

```

interrupts = <0x00000000 0x0000005c 0x00000004 0x00000000 0x0000005d 0x00000004>;
xlnx,gtlane = <0x00000000>;
xlnx,phcindex = <0x00000000>;
max-speed = <0x186A0>;
/* GPIO_ACTIVE_HIGH */
gt_ctrl-gpios = <0x0000005d 0 0>, <0x0000005d 1 0>, <0x0000005d 2 0>, <0x0000005d 3 0>,
<0x0000005d 4 0>, <0x0000005d 5 0>, <0x0000005d 6 0>, <0x0000005d 7 0>,
<0x0000005d 8 0>, <0x0000005d 9 0>, <0x0000005d 10 0>, <0x0000005d 11 0>, <0x0000005d
12 0>, <0x0000005d 13 0>, <0x0000005d 14 0>, <0x0000005d 15 0>,
<0x0000005d 16 0>, <0x0000005d 17 0>, <0x0000005d 18 0>, <0x0000005d 19 0>, <0x000000
5d 20 0>, <0x0000005d 21 0>, <0x0000005d 22 0>, <0x0000005d 23 0>,
<0x0000005d 24 0>, <0x0000005d 25 0>, <0x0000005d 26 0>, <0x0000005d 27 0>, <0x000000
5d 28 0>, <0x0000005d 29 0>, <0x0000005d 30 0>, <0x0000005d 31 0>;

gt_rx_dpath-gpios = <0x0000005e 0 0>, <0x0000005e 1 0>, <0x0000005e 2 0>, <0x0000005e
3 0>, <0x0000005e 4 0>, <0x0000005e 5 0>, <0x0000005e 6 0>, <0x0000005e 7 0>,
<0x0000005e 8 0>, <0x0000005e 9 0>, <0x0000005e 10 0>, <0x0000005e 11 0>, <0x0000005e
12 0>, <0x0000005e 13 0>, <0x0000005e 14 0>, <0x0000005e 15 0>,
<0x0000005e 16 0>, <0x0000005e 17 0>, <0x0000005e 18 0>, <0x0000005e 19 0>, <0x000000
5e 20 0>, <0x0000005e 21 0>, <0x0000005e 22 0>;

gt_tx_dpath-gpios = <0x0000005f 0 0>, <0x0000005f 1 0>, <0x0000005f 2 0>, <0x0000005f
3 0>, <0x0000005f 4 0>, <0x0000005f 5 0>, <0x0000005f 6 0>, <0x0000005f 7 0>,
<0x0000005f 8 0>, <0x0000005f 9 0>, <0x0000005f 10 0>, <0x0000005f 11 0>, <0x0000005f
12 0>, <0x0000005f 13 0>, <0x0000005f 14 0>, <0x0000005f 15 0>,
<0x0000005f 16 0>, <0x0000005f 17 0>, <0x0000005f 18 0>, <0x0000005f 19 0>, <0x000000
5f 20 0>, <0x0000005f 21 0>, <0x0000005f 22 0>;

gt_tx_rst_done-gpios = <0x00000060 0 0>, <0x00000060 1 0>, <0x00000060 2 0>, <0x000000
60 3 0>, <0x00000060 4 0>, <0x00000060 5 0>, <0x00000060 6 0>, <0x00000060 7 0>,
<0x00000060 8 0>, <0x00000060 9 0>, <0x00000060 10 0>, <0x00000060 11 0>, <0x00000060
12 0>, <0x00000060 13 0>, <0x00000060 14 0>, <0x00000060 15 0>,
<0x00000060 16 0>, <0x00000060 17 0>, <0x00000060 18 0>, <0x00000060 19 0>, <0x000000
60 20 0>, <0x00000060 21 0>, <0x00000060 22 0>, <0x00000060 23 0>;

gt_rx_rst_done-gpios = <0x00000060 24 0>, <0x00000060 25 0>, <0x00000060 26 0>, <0x00
000060 27 0>, <0x00000060 28 0>, <0x00000060 29 0>, <0x00000060 30 0>, <0x00000060 31
0>,
<0x00000060 32 0>, <0x00000060 33 0>, <0x00000060 34 0>, <0x00000060 35 0>, <0x000000
60 36 0>, <0x00000060 37 0>, <0x00000060 38 0>, <0x00000060 39 0>,
<0x00000060 40 0>, <0x00000060 41 0>, <0x00000060 42 0>, <0x00000060 43 0>, <0x000000
60 44 0>, <0x00000060 45 0>, <0x00000060 46 0>, <0x00000060 47 0>;

gt_rsts-gpios = <0x00000061 0 0>, <0x00000061 1 0>, <0x00000061 2 0>, <0x00000061 3 0>
, <0x00000061 4 0>, <0x00000061 5 0>, <0x00000061 6 0>, <0x00000061 7 0>,
<0x00000061 8 0>, <0x00000061 9 0>, <0x00000061 10 0>, <0x00000061 11 0>, <0x00000061
12 0>, <0x00000061 13 0>;
xlnx,gt-ch0-rx-user-data-width-c0 = <0x00000050>;
xlnx,gt-mode-1 = "PAM4";
xlnx,gt-ch0-rx-user-data-width-c1 = <0x00000050>;
reg = <0x00000000 0xa4000000 0x00000000 0x00100000>;
xlnx,name = "dcmac_100caui4_1588_core";

```

```

clock-names = "rx_axi_clk", "rx_core_clk", "rx_flexif_clk", "rx_macif_clk",
"s_axi_aclk", "ts_clk", "tx_axi_clk", "tx_core_clk", "tx_flexif_clk", "tx_macif_clk";
xlnx,data-rate-cfg-0 = "100G";
status = "okay";
xlnx,edk-iptype = "PERIPHERAL";
compatible = "xlnx,dcmac-2.4";
xlnx,ip-name = "dcmac";
clocks = <0x000000010 0x000000064 0x000000065 0x000000065 0x000000065 0x000000065
0x000000010 0x000000064 0x000000065 0x000000065>;
xlnx,num-gt-channels = <0x000000004>;
fixed-link {
speed = <100000>;
full-duplex;
};
};

axi_gpio_gt_ctl: gpio@a4130000 {
#gpio-cells = <0x000000002>;
clock-names = "s_axi_aclk";
clocks = <0x000000003 0x000000041>;
compatible = "xlnx,axi-gpio-2.0", "xlnx,xps-gpio-1.00.a";
gpio-controller;
reg = <0x000000000 0xa4130000 0x000000000 0x00010000>;
xlnx,all-inputs = <0x000000000>;
xlnx,all-inputs-2 = <0x000000000>;
xlnx,all-outputs = <0x000000001>;
xlnx,all-outputs-2 = <0x000000000>;
xlnx,dout-default = <0x000000000>;
xlnx,dout-default-2 = <0x000000000>;
xlnx,gpio-width = <0x000000020>;
xlnx,gpio2-width = <0x000000020>;
xlnx,interrupt-present = <0x000000000>;
xlnx,is-dual = <0x000000000>;
xlnx,tri-default = <0xffffffff>;
xlnx,tri-default-2 = <0xffffffff>;
phandle = <0x00000005d>;
};

axi_gpio_rx_datapath: gpio@a4150000 {
#gpio-cells = <0x000000002>;
clock-names = "s_axi_aclk";
clocks = <0x000000003 0x000000041>;
compatible = "xlnx,axi-gpio-2.0", "xlnx,xps-gpio-1.00.a";
gpio-controller;
reg = <0x000000000 0xa4150000 0x000000000 0x00010000>;
xlnx,all-inputs = <0x000000000>;
xlnx,all-inputs-2 = <0x000000000>;
xlnx,all-outputs = <0x000000001>;
xlnx,all-outputs-2 = <0x000000000>;
xlnx,dout-default = <0x000000000>;
xlnx,dout-default-2 = <0x000000000>;
xlnx,gpio-width = <0x000000020>;
xlnx,gpio2-width = <0x000000020>;

```

```

xlnx,interrupt-present = <0x00000000>;
xlnx,is-dual = <0x00000000>;
xlnx,tri-default = <0xffffffff>;
xlnx,tri-default-2 = <0xffffffff>;
phandle = <0x00000005e>;
};
axi_gpio_tx_datapath: gpio@a4140000 {
#gpio-cells = <0x00000002>;
clock-names = "s_axi_aclk";
clocks = <0x00000003 0x00000041>;
compatible = "xlnx,axi-gpio-2.0", "xlnx,xps-gpio-1.00.a";
gpio-controller;
reg = <0x00000000 0xa4140000 0x00000000 0x00010000>;
xlnx,all-inputs = <0x00000000>;
xlnx,all-inputs-2 = <0x00000000>;
xlnx,all-outputs = <0x00000001>;
xlnx,all-outputs-2 = <0x00000000>;
xlnx,dout-default = <0x00000000>;
xlnx,dout-default-2 = <0x00000000>;
xlnx,gpio-width = <0x00000020>;
xlnx,gpio2-width = <0x00000020>;
xlnx,interrupt-present = <0x00000000>;
xlnx,is-dual = <0x00000000>;
xlnx,tri-default = <0xffffffff>;
xlnx,tri-default-2 = <0xffffffff>;
phandle = <0x00000005f>;
};
axi_reset_done_dyn: gpio@a4160000 {
#gpio-cells = <0x00000002>;
clock-names = "s_axi_aclk";
clocks = <0x00000003 0x00000041>;
compatible = "xlnx,axi-gpio-2.0", "xlnx,xps-gpio-1.00.a";
gpio-controller;
reg = <0x00000000 0xa4160000 0x00000000 0x00010000>;
xlnx,all-inputs = <0x00000001>;
xlnx,all-inputs-2 = <0x00000001>;
xlnx,all-outputs = <0x00000000>;
xlnx,all-outputs-2 = <0x00000000>;
xlnx,dout-default = <0x00000000>;
xlnx,dout-default-2 = <0x00000000>;
xlnx,gpio-width = <0x00000018>;
xlnx,gpio2-width = <0x00000018>;
xlnx,interrupt-present = <0x00000000>;
xlnx,is-dual = <0x00000001>;
xlnx,tri-default = <0xffffffff>;
xlnx,tri-default-2 = <0xffffffff>;
phandle = <0x000000060>;
};
axi_resets_dyn: gpio@a4170000 {
#gpio-cells = <0x00000002>;
clock-names = "s_axi_aclk";
clocks = <0x00000003 0x00000041>;

```

```
compatible = "xlnx,axi-gpio-2.0", "xlnx,xps-gpio-1.00.a";
gpio-controller;
reg = <0x00000000 0xa4170000 0x00000000 0x00010000>;
xlnx,all-inputs = <0x00000000>;
xlnx,all-inputs-2 = <0x00000000>;
xlnx,all-outputs = <0x00000001>;
xlnx,all-outputs-2 = <0x00000000>;
xlnx,dout-default = <0x00000000>;
xlnx,dout-default-2 = <0x00000000>;
xlnx,gpio-width = <0x0000000e>;
xlnx,gpio2-width = <0x00000020>;
xlnx,interrupt-present = <0x00000000>;
xlnx,is-dual = <0x00000000>;
xlnx,tri-default = <0xffffffff>;
xlnx,tri-default-2 = <0xffffffff>;
phandle = <0x00000061>;
};
```

- Some PL clocks are missing in "clock-names" and "clocks" properties for XXV node due to which kernel hangs. To fix it, replace both properties with following in XXV node in pl.dtsi:

```
clock-names = "rx_core_clk", "dclk", "s_axi_aclk", "m_axi_mm2s_aclk",
"m_axi_s2mm_aclk", "m_axi_sg_aclk", "s_axi_lite_aclk";
clocks = <&misc_clk_0>, <&zynqmp_clk 72>, <&zynqmp_clk 71>, <&misc_clk_0>,
<&misc_clk_0>, <&zynqmp_clk 71>, <&zynqmp_clk 71>;
```

4.1.1 Important AR links

- AXI Ethernet driver in specific MCDMA configuration throws swiotlb full error with jumbo frames. Please refer to 2020.x [AR-75128](https://www.xilinx.com/support/answers/75218.html)¹.
- Default DTG generation for XXV Ethernet designs fails on 2020.2. Please refer to [AR-76029](https://www.xilinx.com/support/answers/76029.html)² and [AR-76457](https://www.xilinx.com/support/answers/76457.html)³.

¹ <https://www.xilinx.com/support/answers/75218.html>

² <https://www.xilinx.com/support/answers/76029.html>

³ <https://www.xilinx.com/support/answers/76457.html>

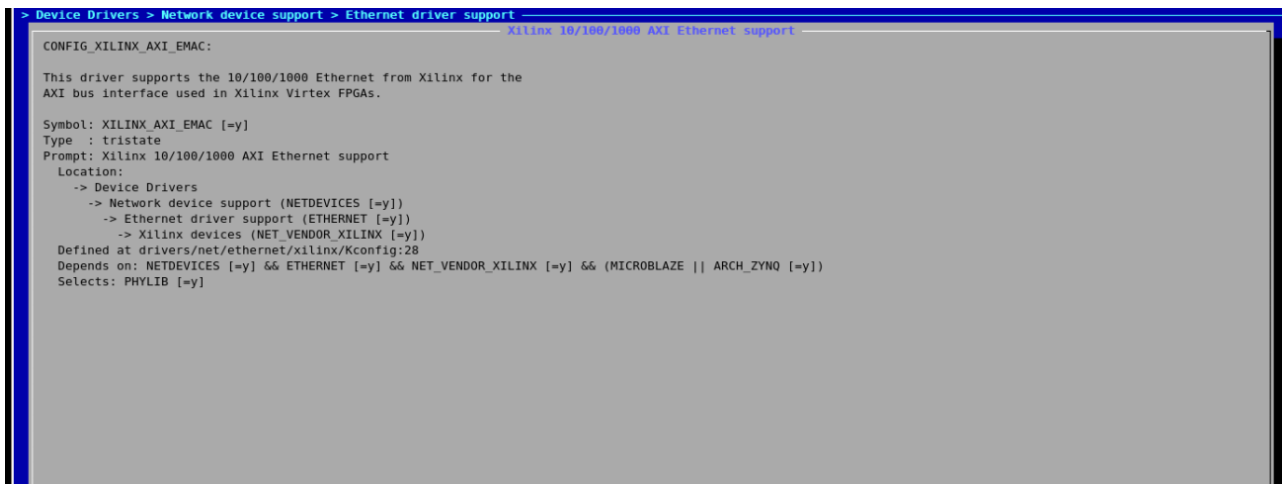
5 Kernel Configuration

The following config options should be enabled in order to build the Axi Ethernet driver

CONFIG_ETHERNET

CONFIG_NET_VENDOR_XILINX

CONFIG_XILINX_AXI_EMAC



```

-> Device Drivers > Network device support > Ethernet driver support
Xilinx 10/100/1000 AXI Ethernet support

CONFIG_XILINX_AXI_EMAC:

This driver supports the 10/100/1000 Ethernet from Xilinx for the
AXI bus interface used in Xilinx Virtex FPGAs.

Symbol: XILINX_AXI_EMAC [=y]
Type : tristate
Prompt: Xilinx 10/100/1000 AXI Ethernet support
Location:
  -> Device Drivers
    -> Network device support (NETDEVICES [=y])
      -> Ethernet driver support (ETHERNET [=y])
        -> Xilinx devices (NET_VENDOR_XILINX [=y])
Defined at drivers/net/ethernet/xilinx/Kconfig:28
Depends on: NETDEVICES [=y] && ETHERNET [=y] && NET_VENDOR_XILINX [=y] && (MICROBLAZE || ARCH_ZYNQ [=y])
Selects: PHYLIB [=y]
  
```

CONFIG_AXIENET_HAS_MCDMA (Select this option In the design if Axi Ethernet is configured with Axi MCDMA)
 CONFIG_XILINX_PHY (For testing SGMII/1000Base-x Configuration with PCS/PMA Core)

6 Device-tree

For mor details on Ethernet bindings, please refer to <https://github.com/Xilinx/linux-xlnx/blob/master/Documentation/devicetree/bindings/net/xlnx%2Caxi-ethernet.yaml>

For more details on phy bindings please refer to <https://github.com/Xilinx/linux-xlnx/blob/master/Documentation/devicetree/bindings/net/ethernet-phy.yaml>

```
axi_ethernet_eth_buf: ethernet@40c00000 {
    axistream-connected = <&axi_dma_1>;
    axistream-control-connected = <&axi_dma_1>;
    clock-frequency = <100000000>;
    clocks = <&clk_bus_0>;
    compatible = "xlnx,axi-ethernet-1.00.a";
    device_type = "network";
    interrupt-parent = <&microblaze_1_axi_intc>;
    interrupts = <4 2>;
    reg = <0x40c00000 0x40000>;
    xlnx,phy-type = <0x4>;
    xlnx,phyaddr = <0x1>;
    xlnx,rxcsun = <0x0>;
    xlnx,rxmem = <0x8000>;
    xlnx,txcsun = <0x0>;
    phy-handle = <&phy0>;
    mdio {
        #address-cells = <1>;
        #size-cells = <0>;
        phy0: phy@7 {
            device_type = "ethernet-phy";
            reg = <7>;
        };
    };
};
```

6.1 Soft Ethernet MAC Configured with MCDMA

6.1.1 (1G, legacy 10G or 10G/25G MAC, MRMAC) with MCDMA

```
xxv_ethernet_0: ethernet@80020000 {
    axistream-connected = <&axi_dma_hier_axi_mcdma_0>;
    axistream-control-connected = <&axi_dma_hier_axi_mcdma_0>;
    clock-frequency = <100000000>;
    clock-names = "rx_core_clk_0", "dclk", "s_axi_aclk_0";
```

```

clocks = <&misc_clk_0>, <&clk_72>, <&clk_71>;
compatible = "xlnx,xxv-ethernet-2.5", "xlnx,xxv-ethernet-1.0";
device_type = "network";
local-mac-address = [00 0a 35 00 00 00];
phy-mode = "base-r";
reg = <0x0 0x80020000 0x0 0x10000>;
xlnx = <0x0>;
xlnx,add-gt-cntrl-sts-ports = <0x0>;
xlnx,anlt-clk-in-mhz = <0x64>;
xlnx,axis-tdata-width = <0x40>;
xlnx,axis-tkeep-width = <0x7>;
xlnx,base-r-kr = "BASE-R";
xlnx,channel-ids = "1","2","3","4","5","6","7","8","9","a","b","c","d","e","f","1
0";
xlnx,clocking = "Asynchronous";
xlnx,core = "Ethernet MAC+PCS/PMA 64-bit";
xlnx,data-path-interface = "AXI Stream";
xlnx,enable-datapath-parity = <0x0>;
xlnx,enable-pipeline-reg = <0x0>;
xlnx,enable-preemption = <0x0>;
xlnx,enable-preemption-fifo = <0x0>;
xlnx,enable-rx-flow-control-logic = <0x0>;
xlnx,enable-time-stamping = <0x1>;
xlnx,enable-tx-flow-control-logic = <0x0>;
xlnx,enable-vlane-adjust-mode = <0x0>;
xlnx,family-chk = "zynqplus";
xlnx,fast-sim-mode = <0x0>;
xlnx,gt-diffctrl-width = <0x4>;
xlnx,gt-drp-clk = "100.00";
xlnx,gt-group-select = "Quad X0Y0";
xlnx,gt-location = <0x1>;
xlnx,gt-ref-clk-freq = "156.25";
xlnx,gt-type = "GTH";
xlnx,include-auto-neg-lt-logic = "None";
xlnx,include-axi4-interface = <0x1>;
xlnx,include-fec-logic = <0x0>;
xlnx,include-rsfec-logic = <0x0>;
xlnx,include-shared-logic = <0x1>;
xlnx,include-user-fifo = <0x1>;
xlnx,lane1-gt-loc = "X0Y4";
xlnx,lane2-gt-loc = "NA";
xlnx,lane3-gt-loc = "NA";
xlnx,lane4-gt-loc = "NA";
xlnx,line-rate = <0xa>;
xlnx,mii-ctrl-width = <0x4>;
xlnx,mii-data-width = <0x20>;
xlnx,num-of-cores = <0x1>;
xlnx,num-queues = /bits/ 16 <0x10>;
xlnx,ptp-clocking-mode = <0x0>;
xlnx,ptp-operation-mode = <0x2>;
xlnx,runtime-switch = <0x0>;
xlnx,rxmem = <0x40000>;

```

```
xlnx_switch-1-10-25g = <0x0>;  
xlnx_tx-latency-adjust = <0x0>;  
xlnx_tx-total-bytes-width = <0x4>;  
xlnx_xgmii-interface = <0x1>;  
interrupt-names = "mm2s_ch1_introut", "mm2s_ch2_introut", "mm2s_ch3_introut",  
"mm2s_ch4_introut", "mm2s_ch5_introut", "mm2s_ch6_introut", "mm2s_ch7_introut",  
"mm2s_ch8_introut", "mm2s_ch9_introut", "mm2s_ch10_introut", "mm2s_ch11_introut",  
"mm2s_ch12_introut", "mm2s_ch13_introut", "mm2s_ch14_introut", "mm2s_ch15_introut",  
"mm2s_ch16_introut", "s2mm_ch1_introut", "s2mm_ch2_introut", "s2mm_ch3_introut",  
"s2mm_ch4_introut", "s2mm_ch5_introut", "s2mm_ch6_introut", "s2mm_ch7_introut",  
"s2mm_ch8_introut", "s2mm_ch9_introut", "s2mm_ch10_introut", "s2mm_ch11_introut",  
"s2mm_ch12_introut", "s2mm_ch13_introut", "s2mm_ch14_introut", "s2mm_ch15_introut",  
"s2mm_ch16_introut";  
interrupt-parent = <&gic>;  
interrupts = <0 89 4 0 89 4 0 89 4 0 89 4 0 89 4 0 89 4 0 89 4 0 89 4 0 89  
4 0 89 4 0 89 4 0 89 4 0 89 4 0 89 4 0 90 4 0 90 4 0 90 4 0 90 4 0 90  
4 0 90 4 0 90 4 0 90 4 0 90 4 0 90 4 0 90 4 0 90 4 0 90 4 0 90 4>;  
  
xxv_ethernet_0_mdio: mdio {  
    #address-cells = <1>;  
    #size-cells = <0>;  
};
```

When Axi Ethernet (10G/25G MAC) configured with MCDMA device-tree node will be like below (to make use of internal MCDMA driver)

6.1.2 DCMAC with MCDMA

When DCMAC (1x100G) is configured with MCDMA, ethernet node will be like below

```
dcmac@a4000000 {
    phy-mode = "100gbase-r";
    xlnx,gt-mode-0 = "NRZ";
    axistream-connected = <0x00000062>;
    xlnx,num-queues = [00 01];
    xlnx,channel-ids = "1";
    interrupt-names = "mm2s_ch1_introut", "s2mm_ch1_introut";
    interrupt-parent = <0x00000005>;
    interrupts = <0x00000000 0x0000005c 0x00000004 0x00000000 0x0000005d 0x00000004>;
    xlnx,gtlane = <0x00000000>;
    xlnx,phcindex = <0x00000000>;
    max-speed = <0x186A0>;
    /* GPIO_ACTIVE_HIGH */
    gt_ctrl-gpios = <0x0000005d 0 0>, <0x0000005d 1 0>, <0x0000005d 2 0>, <0x0000005d
3 0>, <0x0000005d 4 0>, <0x0000005d 5 0>, <0x0000005d 6 0>, <0x0000005d 7 0>,
<0x0000005d 8 0>, <0x0000005d 9 0>, <0x0000005d 10 0>, <0x0000005d 11 0>, <0x0000
005d 12 0>, <0x0000005d 13 0>, <0x0000005d 14 0>, <0x0000005d 15 0>,
<0x0000005d 16 0>, <0x0000005d 17 0>, <0x0000005d 18 0>, <0x0000005d 19 0>, <0x00
00005d 20 0>, <0x0000005d 21 0>, <0x0000005d 22 0>, <0x0000005d 23 0>.
```

```

    <0x00000005d 24 0>, <0x00000005d 25 0>, <0x00000005d 26 0>, <0x00000005d 27 0>, <0x00
00005d 28 0>, <0x00000005d 29 0>, <0x00000005d 30 0>, <0x00000005d 31 0>;

    gt_rx_dpath-gpios = <0x00000005e 0 0>, <0x00000005e 1 0>, <0x00000005e 2 0>, <0x0000
005e 3 0>, <0x00000005e 4 0>, <0x00000005e 5 0>, <0x00000005e 6 0>, <0x00000005e 7 0>,
    <0x00000005e 8 0>, <0x00000005e 9 0>, <0x00000005e 10 0>, <0x00000005e 11 0>, <0x0000
005e 12 0>, <0x00000005e 13 0>, <0x00000005e 14 0>, <0x00000005e 15 0>,
    <0x00000005e 16 0>, <0x00000005e 17 0>, <0x00000005e 18 0>, <0x00000005e 19 0>, <0x00
00005e 20 0>, <0x00000005e 21 0>, <0x00000005e 22 0>;

    gt_tx_dpath-gpios = <0x00000005f 0 0>, <0x00000005f 1 0>, <0x00000005f 2 0>, <0x0000
005f 3 0>, <0x00000005f 4 0>, <0x00000005f 5 0>, <0x00000005f 6 0>, <0x00000005f 7 0>,
    <0x00000005f 8 0>, <0x00000005f 9 0>, <0x00000005f 10 0>, <0x00000005f 11 0>, <0x0000
005f 12 0>, <0x00000005f 13 0>, <0x00000005f 14 0>, <0x00000005f 15 0>,
    <0x00000005f 16 0>, <0x00000005f 17 0>, <0x00000005f 18 0>, <0x00000005f 19 0>, <0x00
00005f 20 0>, <0x00000005f 21 0>, <0x00000005f 22 0>;

    gt_tx_rst_done-gpios = <0x000000060 0 0>, <0x000000060 1 0>, <0x000000060 2 0>, <0x0
0000060 3 0>, <0x000000060 4 0>, <0x000000060 5 0>, <0x000000060 6 0>, <0x000000060 7 0>,
    <0x000000060 8 0>, <0x000000060 9 0>, <0x000000060 10 0>, <0x000000060 11 0>, <0x0000
0060 12 0>, <0x000000060 13 0>, <0x000000060 14 0>, <0x000000060 15 0>,
    <0x000000060 16 0>, <0x000000060 17 0>, <0x000000060 18 0>, <0x000000060 19 0>, <0x00
000060 20 0>, <0x000000060 21 0>, <0x000000060 22 0>, <0x000000060 23 0>;

    gt_rx_rst_done-gpios = <0x000000060 24 0>, <0x000000060 25 0>, <0x000000060 26 0>, <0
x000000060 27 0>, <0x000000060 28 0>, <0x000000060 29 0>, <0x000000060 30 0>, <0x000000060
31 0>,
    <0x000000060 32 0>, <0x000000060 33 0>, <0x000000060 34 0>, <0x000000060 35 0>, <0x00
000060 36 0>, <0x000000060 37 0>, <0x000000060 38 0>, <0x000000060 39 0>,
    <0x000000060 40 0>, <0x000000060 41 0>, <0x000000060 42 0>, <0x000000060 43 0>, <0x00
000060 44 0>, <0x000000060 45 0>, <0x000000060 46 0>, <0x000000060 47 0>;

    gt_rsts-gpios = <0x000000061 0 0>, <0x000000061 1 0>, <0x000000061 2 0>, <0x000000061
3 0>, <0x000000061 4 0>, <0x000000061 5 0>, <0x000000061 6 0>, <0x000000061 7 0>,
    <0x000000061 8 0>, <0x000000061 9 0>, <0x000000061 10 0>, <0x000000061 11 0>, <0x0000
0061 12 0>, <0x000000061 13 0>;
    xlnx,gt-ch0-rx-user-data-width-c0 = <0x000000050>;
    xlnx,gt-mode-1 = "PAM4";
    xlnx,gt-ch0-rx-user-data-width-c1 = <0x000000050>;
    reg = <0x000000000 0xa40000000 0x000000000 0x001000000>;
    xlnx,name = "dcmac_100caui4_1588_core";
    clock-names = "rx_axi_clk", "rx_core_clk", "rx_flexif_clk", "rx_macif_clk",
"s_axi_aclk", "ts_clk", "tx_axi_clk", "tx_core_clk", "tx_flexif_clk", "tx_macif_clk";
    xlnx,data-rate-cfg-0 = "100G";
    status = "okay";
    xlnx,edk-iptype = "PERIPHERAL";
    compatible = "xlnx,dcmac-2.4";
    xlnx,ip-name = "dcmac";
    clocks = <0x000000010 0x000000064 0x000000065 0x000000065 0x000000065 0x000000065
0x000000010 0x000000064 0x000000065 0x000000065>;
    xlnx,num-gt-channels = <0x000000004>;
    fixed-link {

```

```

        speed = <100000>;
        full-duplex;
    };
};

```

When Soft Ethernet MAC configured with MCDMA, The driver supports several features of the MCDMA-

- The driver supports random Queue/Channel selection.

Assume in vivado design MCDMA is configured for 16 channels and user don't want Linux driver to use all the 16 channels.

Below example, pl.dtsi (AXI ethernet node) marks channel 2, 5 and 10 to be used by Linux driver.

```

xlnx,num-queues = <0x3>;
xlnx,channel-ids = "2","5","10";

```

- The driver supports channel observer feature through sysfs. This custom feature is useful in multi-core (Observer) system where MCDMA is a shared resource for all cores. MCDMA IP supports a maximum of six cores and 16 Channels can be distributed across each core as a static configuration. The Channel Observer is available for each group and provides the status about the channels in a group being serviced.
- The driver supports per channel weight configuration through sysfs. This custom feature specifies the channel weight i.e number of packets to be sent in one iteration.
- The driver supports Linux multiqueue networking. It uses the `alloc_etherdev_mq()` function to allocate the subqueues for the device.

The userspace command 'tc,' part of the iproute2 package, is used to configure qdiscs. To add the MULTIQ qdisc assuming the device is called eth0, run the following command:

```
# tc qdisc add dev eth0 root handle 1: multiq
```

The qdisc will allocate the number of bands to equal the number of queues that the device reports, and bring the qdisc online.

Assuming eth0 has 4 Tx queues, the band mapping would look like:

band 0 => queue 0

band 1 => queue 1

band 2 => queue 2

band 3 => queue 3

The behavior of tc filters remains the same. However, a new tc action, skbedit, has been added.

Assuming we want to route all traffic to a specific host, for example 192.168.0.3, through a specific queue we could use this action and establish a filter such as:

```
# tc filter add dev eth0 parent 1: protocol ip prio 1 u32 match ip dst 192.168.0.3 action skbedit
queue_mapping 3
```

For details refer Linux kernel Documentation/networking/multiqueue.txt

To support USXGMII + MCDMA, use above devicetree as reference and select:

```
xlnx,phy-type = <0x7>;
```

6.2 Related device tree information

For PHY related DT information, refer to

<https://github.com/Xilinx/linux-xlnx/blob/master/Documentation/devicetree/bindings/net/ethernet-phy.yaml>

When selecting phy specific settings, make sure to mention interface type, speed (if limited/fixed) and phy address properties.

PHY/Converter devices that may be used with this MAC:

-> Xilinx GMII2RGMII converter (<https://github.com/Xilinx/linux-xlnx/blob/master/Documentation/devicetree/bindings/net/xlnx%2Cgmii-to-rgmii.yaml>)

-> Xilinx PCS PMA PHY (handled internal to this Ethernet driver)

For information on DT bindings when using this Ethernet driver with AXI DMA driver from dma framework, please refer to the following:

See "AXI 1G/2.5G Ethernet Subsystem + AXIDMA without "axistream-connected" property"

<https://github.com/Xilinx/linux-xlnx/blob/master/Documentation/devicetree/bindings/net/xlnx%2Caxi-ethernet.yaml#L268>

7 IEEE 1588 Support

The Existing Axi Ethernet driver in the Xilinx git hub supports 1588 for 1G MAC and legacy 10G MAC and 10G/25G MAC

It does timestamp at the MAC level. 1588 is supported in 7-series and Zynq.

The Current driver assumes that AXI Stream FIFO is connected to the MAC TX Time stamp Stream interface at the design level.

Note:

In order to test the hardware platform, a 1588 timestamp capable timer is required at the h/w level.

Prerequisite for 1588 testing:

Need to enable XILINX_AXI_EMAC_HWTSTAMP in the kernel config which does the timestamping at the MAC level (Tx/Rx)

Users should write their own timer driver for time stamping which does time adjustment/ freq adjustment.

```
axi_ethernet_eth_buf: ethernet@40c00000 {
    axistream-connected = <&axi_dma_1>;
    axififo-connected = <&axi_fifo_0>;
    clock-frequency = <100000000>;
    clocks = <&clk_bus_0>;
    compatible = "xlnx,axi-ethernet-1.00.a";
    device_type = "network";
    interrupt-parent = <&microblaze_1_axi_intc>;
    interrupts = <4 2>;
    reg = <0x40c00000 0x40000>;
    xlnx,phy-type = <0x4>;
    xlnx,phyaddr = <0x1>;
    xlnx,rxcsun = <0x0>;
    xlnx,rxmem = <0x8000>;
    xlnx,txcsun = <0x0>;
    xlnx,txmem = <0x8000>;
    phy-handle = <&phy0>;
    mdio {
        #address-cells = <1>;
        #size-cells = <0>;
        phy0: phy@7 {
            compatible = "marvell,88e1111";
            device_type = "ethernet-phy";
            reg = <7>;
        };
    };
};
```

MRMAC 1588 support is tested with Xilinx 1588 Timer-Syncer block details of which can be found here:

<https://github.com/Xilinx/linux-xlnx/blob/master/Documentation/devicetree/bindings/ptp/ptp-xilinx.yaml>

https://github.com/Xilinx/linux-xlnx/blob/master/drivers/ptp/ptp_xilinx.c

For testing 1588 please refer to Testing tools section below.1588 Device-tree Example Node

7.1 SWITCHABLE 1/10/25G IP

When Axi Ethernet (1G/10G/25G MAC) is configured with DMA device-tree node will be like below

```

axi_dma1: dma@800000000 {
    #dma-cells = <0x01>;
    clock-names =
"s_axi_lite_aclk\0m_axi_sg_aclk\0m_axi_mm2s_aclk\0m_axi_s2mm_aclk";
    clocks = <0x03 0x47 0x03 0x47 0x3b 0x3b>;
    compatible = "not";
    interrupt-names = "mm2s_introut", "s2mm_introut";
    interrupt-parent = <0x04>;
    interrupts = <0x00 0x59 0x04 0x00 0x5a 0x04>;
    reg = <0x00 0x800000000 0x00 0x10000>;
    xlnx,addrwidth = [20];
    xlnx,include-sg;
    xlnx,sg-length-width = <0x10>;
    status = "disabled";
    xlnx,include-dre;
    phandle = <0xac>;

    dma-channel@800000000 {
        compatible = "xlnx,axi-dma-mm2s-channel";
        dma-channels = <0x01>;
        interrupts = <0x00 0x59 0x04>;
        xlnx,datawidth = <0x40>;
        xlnx,device-id = <0x00>;
        xlnx,include-dre;
    };

    dma-channel@800000030 {
        compatible = "xlnx,axi-dma-s2mm-channel";
        dma-channels = <0x01>;
        interrupts = <0x00 0x5a 0x04>;
        xlnx,datawidth = <0x40>;
        xlnx,device-id = <0x00>;
        xlnx,include-dre;
    };
};

ethernet_1_10_25g@80010000 {
    clock-names = "rx_core_clk_0", "rx_core_clk_1", "dclk",
"s_axi_aclk_0", "s_axi_aclk_1";

```

```

clocks = <0x3b 0x3b 0x03 0x48 0x03 0x47 0x03 0x47>;
compatible = "xlnx,ethernet-1-10-25g-2.7";
reg = <0x00 0x80010000 0x00 0x10000>;
xlnx,add-gt-cntrl-sts-ports = <0x00>;
xlnx,anlt-clk-in-mhz = <0x4b>;
xlnx,axis-tdata-width = <0x40>;
xlnx,axis-tkeep-width = <0x07>;
xlnx,base-r-kr = "BASE-R";
xlnx,clocking = "Asynchronous";
xlnx,core = "Ethernet MAC+PCS/PMA 64-bit";
xlnx,data-path-interface = "AXI Stream";
xlnx,enable-pipeline-reg = <0x00>;
xlnx,enable-preemption = <0x00>;
xlnx,enable-rx-flow-control-logic = <0x00>;
xlnx,enable-time-stamping = <0x00>;
xlnx,enable-tx-flow-control-logic = <0x00>;
xlnx,enable-vlane-adjust-mode = <0x00>;
xlnx,family-chk = "zynqplus";
xlnx,fast-sim-mode = <0x00>;
xlnx,gt-drp-clk = "50.00";
xlnx,gt-group-select = "Quad X0Y0";
xlnx,gt-location = <0x01>;
xlnx,gt-ref-clk-freq = "156.25";
xlnx,gt-type = "GTH";
xlnx,include-auto-neg-lt-logic = "Include AN Logic";
xlnx,include-axi4-interface = <0x01>;
xlnx,include-fec-logic = <0x00>;
xlnx,include-lt-logic = "Include LT Logic";
xlnx,include-rsfec-logic = <0x00>;
xlnx,include-shared-logic = <0x01>;
xlnx,include-statistics-counters = <0x01>;
xlnx,include-system-timer-syncers = <0x00>;
xlnx,include-user-fifo = <0x00>;
xlnx,lane1-gt-loc = "X1Y14";
xlnx,lane2-gt-loc = "X1Y15";
xlnx,lane3-gt-loc = "NA";
xlnx,lane4-gt-loc = "NA";
xlnx,line-rate = <0x0a>;
xlnx,num-of-cores = <0x02>;
xlnx,ptp-clocking-mode = <0x00>;
xlnx,ptp-operation-mode = <0x02>;
xlnx,runtime-switch = "1G / 10G";
xlnx,statistics-counters-size = <0x30>;
xlnx,statistics-regs-type = <0x00>;
xlnx,timer-format = <0x00>;
xlnx,tx-latency-adjust = <0x00>;
xlnx,tx-total-bytes-width = <0x04>;
xlnx,xgmii-interface = <0x01>;
axistream-connected = <&axi_dma1>;
axistream-control-connected = <&axi_dma1>;
local-mac-address = [00 0a 35 00 00 00];
interrupt-parent = <0x04>;
interrupt-names = "mm2s_introut", "s2mm_introut";

```

```
interrupts = <0x00 0x59 0x04 0x00 0x5a 0x04>;  
xlnx,addrwidth = [20];  
phandle = <0xae>;  
};
```

8 Performance

These benchmark performance numbers were obtained by connecting Xilinx boards to Linux PCs/server machines (Ubuntu/Red Hat Enterprise).

The tools used are netperf or iperf (Refer to tool information below).

Performance benchmark numbers mentioned in below tables are for reference and dependent on multiple factors i.e setup , vivado design configuration etc.

NOTE: CPU utilization reported in below performance tables is an aggregate of all CPU's. i.e on ZynqMP platform, it reports combined utilization of all four A53 cores.

8.1 1G Ethernet with AXIDMA

Kernel version: 6.1

AXI 1G/2.5G Ethernet Subsystem : TX and RX full checksum offload enabled.

ZynqMP

Board: ZCU102 board (production silicon) + SFP Module

Note: Performance numbers with v6.6 linux are lower than reported here, we will attach AR with fix for that soon.

	TCP (Mbps)				UDP (Mbps)			
MTU	TX	CPU(%)	RX	CPU(%)	TX	CPU(%)	RX	CPU(%)
1500	941	12.78	843	46.79	957	21.13	951	51.09
8192	989	2.34	989	7.26	779	2.98	992	5.86

Zynq

Board: ZC706 board + SFP Module

NOTE- There is ~10% drop (compared to 2019.2) in performance for 1500 MTU.

The drop is due to enable CONFIG_OPTIMIZE_INLINING forcibly" [commit](#)⁴in linux kernel.

Kernel and networking stack is full of inline functions and it could be some unoptimized inline function (could also be dependent on gcc version) leading to a performance drop.

The performance drop is observed on GEM and Xilinx Axi Ethernet MAC's on Zynq

The plan is to document the performance drop on zynq and initiate the discussion with the mainline community so that it is analyzed by respective kernel maintainers.

	TCP (Mbps)	UDP (Mbps)
--	------------	------------

⁴ <https://github.com/Xilinx/linux-xlnx/commit/ac7c3e4ff401b304489a031938dbeaab585bfe0a>

MTU	TX	CPU(%)	RX	CPU(%)	TX	CPU(%)	RX	CPU(%)
1500	740	67.53	537	89.39	453	52.86	456	88.72
8192	977	60.69	732	50.26	743	36.10	643	50.32

Microblaze

Traditionally microblaze designs are not targeted for high performance applications so only functional sanity is done.

8.2 10G Ethernet with AXIMCDMA

ZynqMP

Board: ZCU102 board (production silicon) + SFP Module

	TCP (Gbps)				UDP (Gbps)			
MTU	TX	CPU(%)	RX	CPU(%)	TX	CPU(%)	RX	CPU(%)
1500	2.29	50.91	1.76	64.71	3.03	99.93	1.65	70.04
9000	5.0	52.35	3.56	53.9	6.51	65.09	4.63	54.28

NOTE**Setup Details**

Host setup: Dell System Precision Tower 7910 (0619)

Iperf:

OS : Ubuntu 22.04.2 LTS (Linux kernel 5.15.0-87-generic)

NIC (10G Solarflare's SFN6322F Dual-Port 10GbE SFP+ Adapter) : Default

Performance benchmarking

Kernel version: 6.6: In this design 1588 is not enabled. iperf 3-CURRENT (cJSON 1.5.2)

NOTE: Better performance numbers are observed with linux v6.6 and flag -P 2 in iperf command.

Pre-requisites:

- **Set Ethernet MCDMA TX interrupt affinity to core-1**

```
root@10g-mcdma-no1588-build:~# echo 2 > /proc/irq/xx/smp_affinity
```

- **Run iperf servers on ZynqMP (core2 and core3)**

```
root@10g-mcdma-no1588-build:~# taskset -c 2 iperf3 -s -p 5101 &
```

```
root@10g-mcdma-no1588-build:~# taskset -c 3 iperf3 -s -p 5102 &
```

- **CPU Utilization reporting**

```
root@10g-mcdma-no1588-build:~# ./mpstat -P ALL 1 50
```

- **Run iperf servers on the remote host**

```
server:~# iperf3 -s -p 5101 & ; iperf3 -s -p 5102 & ; iperf3 -s -p 5103 & ; iperf3 -s -p 5104 &
```

Steps:

[illegible]

Test Procedure

8.3 Diagnostic and Protocol Tests

PING

This utility used to test the reachability of a host on an Internet Protocol(IP) network and to measure the round trip time for messages sent from the originating host to a destination computer

How to Run

ping <Remote IP Address>

Telnet

```
telnet <Server IP Address>
```

Pkt Generator

Please refer the below link for how to run and various options

<https://www.kernel.org/doc/Documentation/networking/pktgen.txt>

8.4 Stress Test

Iperf with option -d

Run iperf in dual testing mode. This will cause the server to connect back to the client on the port specified in the `-L5` option (or defaults to the port the client connected to the server on). This is done immediately therefore running the tests simultaneously.

```
./iperf -c <Server IP> -d
```

Ping flood test

Users can send hundred or more packets per second using `-f` option. It prints a `'.'` when a packet is sent, and a backspace is printed when a packet is received

```
ping -f localhost
```

8.5 Performance Test

Netperf

More information please refer to the below link

<http://www.netperf.org/netperf/>

How to Run

Server:

```
netserver
```

Client:

```
taskset 2 ./netperf -H <Server IP> -t TCP_STREAM
```

```
taskset 2 ./netperf -H <Server IP> -t UDP_STREAM
```

Iperf

More information please refer to the below link

<http://en.wikipedia.org/wiki/Iperf>

How to Run

Server:

```
./iperf -s -u
```

```
./iperf -s
```

Client:

```
./iperf -c <Server IP> -u -b <bandwidth>
```

```
./iperf -c <Server IP>
```

⁵ <http://iperf.fr/#listenport>

8.6 1588 Testing

ptp4l implements the PTP boundary clock and ordinary clock. When hardware time stamping is enabled, ptp4l synchronizes the PTP hardware clock to the master clock. phc2sys can be used to synchronize the system clock to the PTP hardware clock on the network interface card (NIC).

Build Instructions for ptp4l application

- Get the source

```
git clone http://git.code.sf.net/p/linuxptp/code linuxptp
```

- Set the CROSS_COMPILE environment variable arm toolchain
- Install the kernel headers

https://www.kernel.org/doc/Documentation/kbuild/headers_install.txt

- Include the headers path in makefile

```
INC = -I/proj/epdsw/punnaiah/git/test/ethernet/1588/header/include
CFLAGS = -Wall $(VER) $(incdefs) $(DEBUG) $(INC) $(EXTRA_CFLAGS)
```

- run make

Execution steps

In order to perform master-slave sync, run the following:

Master (linux server) : ptp4l -i <interface name> -m

Slave (xilinx board) : ptp4l -i <interface name> -m -s

NOTE: If intended before synchronization phc2sys -s <devicename> -w & can be run to synchronize the system clock to a PTP hardware clock.

```
root@xxv1588:~# ptp4l -i eth0 -m -s
ptp4l[58.954]: selected best master clock 000a35.ffff.002201
ptp4l[60.561]: port 1: new foreign master 000f53.ffff.0e9fb0-1
ptp4l[64.560]: selected best master clock 000f53.ffff.0e9fb0
ptp4l[64.560]: running in a temporal vortex
ptp4l[64.560]: port 1: LISTENING to UNCALIBRATED on RS_SLAVE
ptp4l[66.560]: master offset -1474033842509728292 s0 freq +0 path delay 661
ptp4l[67.560]: master offset -1474033842509733149 s1 freq -4857 path delay 1379
ptp4l[68.560]: master offset 622 s2 freq -4235 path delay 1379
ptp4l[68.560]: port 1: UNCALIBRATED to SLAVE on MASTER_CLOCK_SELECTED
ptp4l[69.560]: master offset 536 s2 freq -4134 path delay 1379
```

Synchronization is stabilized in a few secs.

```
ptp4l[90.558]: master offset      9 s2 freq -4306 path delay      87
ptp4l[91.557]: master offset    166 s2 freq -4147 path delay      45
ptp4l[92.557]: master offset     34 s2 freq -4229 path delay      45
ptp4l[93.557]: master offset   -104 s2 freq -4357 path delay      45
ptp4l[94.557]: master offset     -1 s2 freq -4285 path delay      54
```

9

Mainline status

The current Axi Ethernet driver is not in sync with mainline. The following components need to be upstreamed:

- NAPI Support
- Support for 1588
- Support for legacy 10G MAC
- Support for 1G MAC Non-processor Mode
- Support for 10G/25G MAC with 1588
- Support for 2.5G MAC with 1588
- Support for Axi Ethernet with AXI MCDMA Configuration
- Support for USXGMII IP
- Support for MRMAC IP
- Support for 1G/10G/25G IP
- Support for DCMAC IP

Also note that current driver has support for both Ethernet with internal DMA driver and external DMA engine driver.

10 Change Log

2024.2

- Phylink support for 1G/10G/25G IP
- Deprecate "xlnx,mrmac" and use generic "max-speed" DT property.
- DCMAC support
- Use single MAC type instead of two different MAC types for AXI 1G/2.5G ethernet.

<https://github.com/Xilinx/linux-xlnx/commits/xilinx-v2024.2/drivers/net/ethernet/xilinx>

2024.1

- 6.6 kernel rebase.
- Added phylink support for XXV.
- Minor bugfix related to buffer size when DRE disabled.

<https://github.com/Xilinx/linux-xlnx/commits/xilinx-v2024.1/drivers/net/ethernet/xilinx>

2023.2

- Acquire ptp device information dynamically
- Add switchable 1/10/25G MAC support
- Implement work queue to enable/disable link training.

Commits

<https://github.com/Xilinx/linux-xlnx/commits/xilinx-v2023.2/drivers/net/ethernet/xilinx>

2023.1

- 6.1 kernel rebase
- Added phylink support

Commits:

<https://github.com/Xilinx/linux-xlnx/commits/xilinx-v2023.1/drivers/net/ethernet/xilinx>

2022.2

- Bugfix on TX TS FIFO read sequence
- Upstream fixes for RX ring refill allocation failure handling and removing dependency on phy-node for axienet_mdio_setup function call
- Minor fixes on code indentation and mixed declarations.

2022.1

- 5.15 kernel rebase
- MRMAC bugfix to reporting correct link status at 10G/25G.
- XXV bugfix to check for new IP register reporting clock stability.
- Bugfixes for DMA error path handling.
- Minor cleanup for documentation and GCC warnings.

- Note: TSN driver support moved to staging - no effect on any of the Ethernet components discussed in this page.

<https://github.com/Xilinx/linux-xlnx/commits/xilinx-v2022.1/drivers/net/ethernet/xilinx>

2021.2

- Correct phy-mode handling and pass correct mode to PHY framework.
- Fix MCMDA crash due to dma map and memory errors by handling the same.
- Emaclite bug fixes

<https://github.com/Xilinx/linux-xlnx/commits/master/drivers/net/ethernet/xilinx>

https://github.com/Xilinx/linux-xlnx/commits/xlnx_rebase_v5.10_2021.2/drivers/net/ethernet/xilinx

2021.1

- 5.10 kernel rebase
- MRMAC one step support
- Ethtool stats support

[0605a36e0574](#)⁶net: xilinx: Ethtool statistics support

[21d0b2896f3f](#)⁷net: xilinx: axiethernet: Fix compilation error when HW timestamp is enabled

[9ee7bae04b55](#)⁸net: axienet: MRMAC reset sequence fixes

[8a885d4a28ef](#)⁹net: axienet: Add one step PDelay support

[96bdccafe01a](#)¹⁰net: xilinx: Add one step 1588 support for MRMAC

[b9467dc270a2](#)¹¹net: xilinx: MRMAC TX CMD FIFO update

[365e87b2465e](#)¹²net: xilinx: Handle TX TS FIFO timeout error

[c155fed4cc52](#)¹³net: xilinx: Update MRMAC config structure with PTP command length

2020.2

-
- 6 <https://github.com/Xilinx/linux-xlnx/commit/0605a36e057480f3a83ae401e7ff59739da78e82#diff-5195fe056ef2208d4ea59617c72d8eb157fce3184f1f62fb277d342ccda99749>
- 7 <https://github.com/Xilinx/linux-xlnx/commit/21d0b2896f3f1aa404631f7c89620431e4f3e021#diff-5195fe056ef2208d4ea59617c72d8eb157fce3184f1f62fb277d342ccda99749>
- 8 <https://github.com/Xilinx/linux-xlnx/commit/9ee7bae04b55247e9610720259887cad6759495b#diff-5195fe056ef2208d4ea59617c72d8eb157fce3184f1f62fb277d342ccda99749>
- 9 <https://github.com/Xilinx/linux-xlnx/commit/8a885d4a28ef180a86c0228a93369903cbb8af7f#diff-5195fe056ef2208d4ea59617c72d8eb157fce3184f1f62fb277d342ccda99749>
- 10 <https://github.com/Xilinx/linux-xlnx/commit/96bdccafe01a32c2e8ef8782f83e2afa40b8c320#diff-5195fe056ef2208d4ea59617c72d8eb157fce3184f1f62fb277d342ccda99749>
- 11 <https://github.com/Xilinx/linux-xlnx/commit/b9467dc270a2a57ed35b2ba49ed0125a42b5c065#diff-5195fe056ef2208d4ea59617c72d8eb157fce3184f1f62fb277d342ccda99749>
- 12 <https://github.com/Xilinx/linux-xlnx/commit/365e87b2465e57db5676d5e9adc6d0c4ea0b6933#diff-5195fe056ef2208d4ea59617c72d8eb157fce3184f1f62fb277d342ccda99749>
- 13 <https://github.com/Xilinx/linux-xlnx/commit/c155fed4cc52d85beb1a477054136416be088bdd#diff-5195fe056ef2208d4ea59617c72d8eb157fce3184f1f62fb277d342ccda99749>

- Fix kernel crash on running high traffic in 10G MCDMA PTP design
- Add MRMAC support and related ptp fixes
- Pass phy interface to of_phy_connect()
- Fix swiotlb buffer full error
- In _mcdma.c check if dma_map_single returned a valid address

[e966fbb13383](#)¹⁴net: xilinx: axiethernet: Add check for transmit data FIFO vacancy

[2c44afef9055](#)¹⁵net: xilinx: Fix indentation issues

[664e0cc2775f](#)¹⁶net: xilinx: Add HW timestamping support for MRMAC

[9036464818ee](#)¹⁷net: xilinx: Add MRMAC support

[b83c55d11cbd](#)¹⁸net: xilinx: XXV Ethernet HW timestamp fixes

[e693fba20632](#)¹⁹net: xilinx: Use prandom with range to avoid corner cases

[4818d3994645](#)²⁰net: xilinx: axiethernet: Pass phy interface to of_phy_connect()

[0245a7128384](#)²¹net: xilinx: axiethernet: Fix swiotlb buffer full error

[8b8ae289f6d3](#)²²net: xilinx: Check if dma_map_single returned a valid address.

2020.1

- Kernel upgrade 5.4 and rebase existing axiethernet patches on top of it.

2019.2

- Regression fixes i.e crash in axienet_recv , kernel bug in ndo_open and 1588.
- Enhance error and debug reporting.

Commit IDs:

¹⁴<https://github.com/Xilinx/linux-xlnx/commit/e966fbb13383e0cf1d1ab444b248bd0653d2c05#diff-a1bfa446162b85e07ff5f27cf9ee06664d8c6d55c68e236da5019bf71589bff2>

¹⁵<https://github.com/Xilinx/linux-xlnx/commit/2c44afef9055d6209aed72e81ce357e49ae755ed#diff-a1bfa446162b85e07ff5f27cf9ee06664d8c6d55c68e236da5019bf71589bff2>

¹⁶<https://github.com/Xilinx/linux-xlnx/commit/664e0cc2775faf252f4a55d58bb30f622cc7cddf#diff-a1bfa446162b85e07ff5f27cf9ee06664d8c6d55c68e236da5019bf71589bff2>

¹⁷<https://github.com/Xilinx/linux-xlnx/commit/9036464818ee418e2a212ba2d13f16c197617050#diff-a1bfa446162b85e07ff5f27cf9ee06664d8c6d55c68e236da5019bf71589bff2>

¹⁸<https://github.com/Xilinx/linux-xlnx/commit/b83c55d11cbd986418edbf4b407ab0c8bbe2efb8#diff-a1bfa446162b85e07ff5f27cf9ee06664d8c6d55c68e236da5019bf71589bff2>

¹⁹<https://github.com/Xilinx/linux-xlnx/commit/e693fba206326fc8e45a9ddfc933ffe8dfe830a8#diff-a1bfa446162b85e07ff5f27cf9ee06664d8c6d55c68e236da5019bf71589bff2>

²⁰<https://github.com/Xilinx/linux-xlnx/commit/4818d39946450ce201729671d91d6c1b23b9974d#diff-a1bfa446162b85e07ff5f27cf9ee06664d8c6d55c68e236da5019bf71589bff2>

²¹<https://github.com/Xilinx/linux-xlnx/commit/0245a7128384583d6c0d934dec00d45a9257903c#diff-a1bfa446162b85e07ff5f27cf9ee06664d8c6d55c68e236da5019bf71589bff2>

²²<https://github.com/Xilinx/linux-xlnx/commit/8b8ae289f6d34559b47dbf81f65077ad398f87d5#diff-ed3db2962641eb6ab0bdb0a4c5865f2419901d681fa86015889d6f194ffd3561>

[be66caf](#)²³ net: xilinx: axiethernet: Add error reporting for DMA probe
[5917934](#)²⁴ net: xilinx: axiethernet: Remove clk_init error message for probe defer
[1076feb](#)²⁵ net: xilinx: axiethernet: Fix kernel bug in ndo_open
[fb9853e](#)²⁶ net: xilinx: axiethernet: Fix crash in axienet_recv
[3aecd4f](#)²⁷ net: xilinx: axiethernet: Fix xxv mac short frame handling
[7069a2f](#)²⁸ net: xilinx: axiethernet: Fixed dev_info message
[725b6e3](#)²⁹ net: xilinx: axiethernet: Add debug messages for TX timestamp
[8f09da4](#)³⁰ net: xilinx: axiethernet: Fix xxv mac tx timestamp
[9c0e07f](#)³¹ net: xilinx: axiethernet: In axienet_device_reset add missing txs fifo reset

2019.1

- Fix crash in ifconfig down
- Fix axiethernet register description
- Check for queue full in transmit path
- Add 64-bit support
- Extend clocking support
- Fix kernel crash on MII ioctl
- use channel-id for mcdma interrupt names
- Fix netconsole implementation
- Trivial code cleanup

Commit IDs:

23 <https://github.com/Xilinx/linux-xlnx/commit/be66cafc97441b597b8de73b5c0a0c775c6ada14#diff-79b4c5af77f83bc2aab258db04000161>
 24 <https://github.com/Xilinx/linux-xlnx/commit/59179348553620da1d9762e5ca2b785cd4c3710a#diff-79b4c5af77f83bc2aab258db04000161>
 25 <https://github.com/Xilinx/linux-xlnx/commit/1076feb13c0cfc89d9ac966da5b02f9a36842fcd#diff-79b4c5af77f83bc2aab258db04000161>
 26 <https://github.com/Xilinx/linux-xlnx/commit/fb9853e39e732395c3896470197cfb3114c730ec#diff-79b4c5af77f83bc2aab258db04000161>
 27 <https://github.com/Xilinx/linux-xlnx/commit/3aecd4f8e2dc87d468559730e143f9e37b6083a0#diff-79b4c5af77f83bc2aab258db04000161>
 28 <https://github.com/Xilinx/linux-xlnx/commit/7069a2ff7cd275d9dfb0596d1146f7ce09f526ab#diff-79b4c5af77f83bc2aab258db04000161>
 29 <https://github.com/Xilinx/linux-xlnx/commit/725b6e367a4e50250458f6102d86aafd8b37cbe5#diff-79b4c5af77f83bc2aab258db04000161>
 30 <https://github.com/Xilinx/linux-xlnx/commit/8f09da4c85b76c3616723f94660ebe4408c517c9#diff-79b4c5af77f83bc2aab258db04000161>
 31 <https://github.com/Xilinx/linux-xlnx/commit/9c0e07f92f6f9248c069867a4e89bffb663c615#diff-79b4c5af77f83bc2aab258db04000161>

[33ebfdb](#)³² net: xilinx: axiethernet: Fix crash in ifconfig down
[f5b9e58](#)³³ net: xilinx: axiethernet: Fix axiethernet register description
[e491e78](#)³⁴ net: xilinx: axiethernet: Check for queue full in transmit path
[0ba2b93](#)³⁵ net: xilinx: axiethernet: Fix code checker warnings
[d4c6c09](#)³⁶ net: xilinx: axiethernet: Use %pa format specifier for phys_addr_t type
[270968c](#)³⁷ net: xilinx: axiethernet: Add 64-bit support
[d139077](#)³⁸ net: xilinx: axiethernet: Extend clocking support
[fdce589](#)³⁹ net: xilinx: axiethernet: Fix kernel crash on MII ioctl
[3f2d6cd](#)⁴⁰ net: xilinx: axiethernet: use channel-id for mcdma interrupt names
[aaad9c0](#)⁴¹ net: xilinx: axiethernet: Fix netconsole implementation

2018.3

- Sync kconfig description.
- In `axienet_skb_tstsmpt()` failure return `TX_BUSY`.
- Add error output on DMA allocation failed.
- Fix memory leak in `axienet_bd_free()`.
- Refactor and split `axidma` and `mcdma` programming in separate sources.
- Fix `dma` name buffer size and `skb_free` in `xmit`.
- Format `XXV` error output.
- Fix compiler warnings.

Commit IDs:

[462eb9d](#)⁴² net: xilinx: Sync kconfig description
[fbb56dc](#)⁴³ net: xilinx: axiethernet: Move `axienet_bd_free()` to `xilinx_axienet_dma.c`
[6d3669b](#)⁴⁴ net: xilinx: axiethernet: Refactor `mcdma_tx/rx_probes`
[83241b6](#)⁴⁵ net: xilinx: axiethernet: Reordered `ethtool` status to print all Tx & Rx stats

32 <https://github.com/Xilinx/linux-xlnx/commit/33ebfdbb0b9f3b6a6d6e9f9c51aa63e10716539c#diff-79b4c5af77f83bc2aab258db04000161>
 33 <https://github.com/Xilinx/linux-xlnx/commit/f5b9e58a3544363b98021284beb44937d8a1cfd0#diff-79b4c5af77f83bc2aab258db04000161>
 34 <https://github.com/Xilinx/linux-xlnx/commit/e491e788cc8868c0a736f2923c957b210067cca7#diff-79b4c5af77f83bc2aab258db04000161>
 35 <https://github.com/Xilinx/linux-xlnx/commit/0ba2b9308b83956da67b8b38d9abd8731c2ec057#diff-79b4c5af77f83bc2aab258db04000161>
 36 <https://github.com/Xilinx/linux-xlnx/commit/d4c6c09d7f6138e461a16e834109c2acd5de02d2#diff-3b1e22b744b9477bc3ba2bd172108c9f>
 37 <https://github.com/Xilinx/linux-xlnx/commit/270968c7fd09be31644344f14cb35ff607b47592#diff-79b4c5af77f83bc2aab258db04000161>
 38 <https://github.com/Xilinx/linux-xlnx/commit/d13907706044d27d328fcf7ff861f29b16629444#diff-79b4c5af77f83bc2aab258db04000161>
 39 <https://github.com/Xilinx/linux-xlnx/commit/fdce58951579c68875e036a6a87f579ea4da2d05#diff-79b4c5af77f83bc2aab258db04000161>
 40 <https://github.com/Xilinx/linux-xlnx/commit/3f2d6cd5a6b14fb191af26ffd811ec90d268786c#diff-3b1e22b744b9477bc3ba2bd172108c9f>
 41 <https://github.com/Xilinx/linux-xlnx/commit/aaad9c0b7a1c24adaff7971a8885b27288d83a5b#diff-79b4c5af77f83bc2aab258db04000161>
 42 <https://github.com/Xilinx/linux-xlnx/commit/462eb9df49c8ecfec5f1ff1cd28c448eee06ad9e#diff-3b1e22b744b9477bc3ba2bd172108c9f>
 43 <https://github.com/Xilinx/linux-xlnx/commit/fbb56dc7e2f25033c72d93356090f849b7b8b49d#diff-3b1e22b744b9477bc3ba2bd172108c9f>
 44 <https://github.com/Xilinx/linux-xlnx/commit/6d3669b537e4a3320809aaa23488cde62a7b4151#diff-3b1e22b744b9477bc3ba2bd172108c9f>
 45 <https://github.com/Xilinx/linux-xlnx/commit/83241b644a9cf38e080c2ea94b5e53ac1b036aa7#diff-3b1e22b744b9477bc3ba2bd172108c9f>

- [1fd503a](#)⁴⁶ net: xilinx: axiethernet: Have separate num_queues for tx and rx queues
- [7758b67](#)⁴⁷ net: xilinx: axiethernet: In axienet_skb_tstsmpt() correct return NETDEV_TX_BUSY on failure
- [7107ea5](#)⁴⁸ net: xilinx: axiethernet: Add error output on DMA allocation failed
- [f16fc2d](#)⁴⁹ net: xilinx: axiethernet: Fix memory leak in axienet_bd_free()
- [528a9db](#)⁵⁰ net: xilinx: axiethernet: Split AXIMCDMA implementation in separate file
- [7169681](#)⁵¹ net: xilinx: axiethernet: Split AXIDMA implementation in separate file
- [8e2082e](#)⁵² net: xilinx: axiethernet: Correct skb free function
- [3e1c99a](#)⁵³ net: xilinx: axiethernet: Correct dma name buffer size
- [aa644d2](#)⁵⁴ net: xilinx: axiethernet: Format XXV MAC error output
- [dc81819](#)⁵⁵ net: xilinx: axiethernet: Fix compiler warnings
- [127d458](#)⁵⁶ net: xilinx: axiethernet: Fix compiler warnings

2018.2

- None

2018.1

- Fix xxv mac padding issue.
- Add support for USXGMII IP.
- Update interrupt-names property.
- Code cleanup of tx with no DRE.
- Fix remove() bug for XXV no MDIO.

Commit IDs:

- [10681b8](#)⁵⁷ net: ethernet: xilinx: update interrupt-names property with ip interrupt naming convention

⁴⁶ <https://github.com/Xilinx/linux-xlnx/commit/1fd503aeadd13e95586a919a1f92f0848ac252acf#diff-3b1e22b744b9477bc3ba2bd172108c9f>

⁴⁷ <https://github.com/Xilinx/linux-xlnx/commit/7758b6707e58bb31f66370db88fd7e7e82f7bfd6#diff-3b1e22b744b9477bc3ba2bd172108c9f>

⁴⁸ <https://github.com/Xilinx/linux-xlnx/commit/7107ea5de33c09099f678346af8067fd4570c1f7#diff-3b1e22b744b9477bc3ba2bd172108c9f>

⁴⁹ <https://github.com/Xilinx/linux-xlnx/commit/f16fc2db609b2a25a3a2500c9eefa9fe64777743#diff-3b1e22b744b9477bc3ba2bd172108c9f>

⁵⁰ <https://github.com/Xilinx/linux-xlnx/commit/528a9db7f9347c8e2936174ef568c0fa6191d555#diff-3b1e22b744b9477bc3ba2bd172108c9f>

⁵¹ <https://github.com/Xilinx/linux-xlnx/commit/71696811cb3f3c1e8c1ab40ae8f2f72de89363ba#diff-3b1e22b744b9477bc3ba2bd172108c9f>

⁵² <https://github.com/Xilinx/linux-xlnx/commit/8e2082e07c4e837ebe21d51a8c4eae17cbe30406#diff-3b1e22b744b9477bc3ba2bd172108c9f>

⁵³ <https://github.com/Xilinx/linux-xlnx/commit/3e1c99aa9e1ec02b7e4ca6c9d87ac08959cd0621#diff-3b1e22b744b9477bc3ba2bd172108c9f>

⁵⁴ <https://github.com/Xilinx/linux-xlnx/commit/aa644d284141163349d39637f1e681786aa7e7a8#diff-3b1e22b744b9477bc3ba2bd172108c9f>

⁵⁵ <https://github.com/Xilinx/linux-xlnx/commit/dc818196c3287582b78959671ea31fbd52e36a53#diff-3b1e22b744b9477bc3ba2bd172108c9f>

⁵⁶ <https://github.com/Xilinx/linux-xlnx/commit/127d458a2104fd11cd9e0b769342cbccca154263#diff-3b1e22b744b9477bc3ba2bd172108c9f>

⁵⁷ <https://github.com/Xilinx/linux-xlnx/commit/10681b8483c193fa00ce22bb54671e4549689a58#diff-79b4c5af77f83bc2aab258db04000161>

[fe44c16](#)⁵⁸ net: ethernet: xilinx: Fix xxv mac padding issue - only pad last element.
[cea5c97](#)⁵⁹ net: ethernet: xilinx: axienet cleanup of tx with no dre
[f5de0cd](#)⁶⁰ net: ethernet: xilinx: only teardown mdio when available
[9733c76](#)⁶¹ net: xilinx: axiethernet: Add USXGMII support

2017.4

- None

2017.3

- Added Support for Ethernet MCDMA Configuration
- Added Support for 1588 in buffered mode configuration.
- Fixed race condition in the transmit path
- Fixed race condition in the random queue selection for Ethernet MCDMA configuration

Commit Id's:

[2108c9f](#)⁶² net: ethernet: xilinx: Add support for mcdma
[2108c9f](#)⁶³ net: ethernet: xilinx: Add support for 1588 in buffered mode
[2108c9f](#)⁶⁴ net: ethernet: xilinx: Fix race condition in the tx path
[2108c9f](#)⁶⁵ net: ethernet: xilinx: Fix race in the random queue selection

2017.2

- None

2017.1

- Added Support for 10G/25G MAC (PG210)
- Added Support for 2.5G MAC
- Fixed issues with the without DRE DMA based design on zynqMP SOC
- Added Clock Support
- Fixed bug in the rx reject interrupt handling.

⁵⁸ <https://github.com/Xilinx/linux-xlnx/commit/fe44c16b943999f989abe5dcf8335d6a4e3d6104#diff-79b4c5af77f83bc2aab258db04000161>
⁵⁹ <https://github.com/Xilinx/linux-xlnx/commit/cea5c97e8d37e3b26b2894bd488228790904ac86#diff-79b4c5af77f83bc2aab258db04000161>
⁶⁰ <https://github.com/Xilinx/linux-xlnx/commit/f5de0cd44415bc6eb7567cb3ae3488eb518f2afc#diff-79b4c5af77f83bc2aab258db04000161>
⁶¹ <https://github.com/Xilinx/linux-xlnx/commit/9733c7684a2552421b7c46eef448f2c9696c157#diff-3b1e22b744b9477bc3ba2bd172108c9f>
⁶² <https://github.com/Xilinx/linux-xlnx/commit/097d723271f3103892f7c6eee31865ec94dd0abf#diff-3b1e22b744b9477bc3ba2bd172108c9f>
⁶³ <https://github.com/Xilinx/linux-xlnx/commit/d44b130dd9b1686b09ce444408ebfd82719a991d#diff-3b1e22b744b9477bc3ba2bd172108c9f>
⁶⁴ <https://github.com/Xilinx/linux-xlnx/commit/ee40c7e77f9b3935b1b88b19345644cec1e1242b#diff-3b1e22b744b9477bc3ba2bd172108c9f>
⁶⁵ <https://github.com/Xilinx/linux-xlnx/commit/8adf435b58087eb6cbce59141c51a0851b43c03a#diff-3b1e22b744b9477bc3ba2bd172108c9f>

Commit Id's:

- [22fa015](#)⁶⁶ net: ethernet : xilinx Add config structure to different axienet macs
- [e3c3b23](#)⁶⁷ net: ethernet: Add quirk checks in the driver
- [453042c](#)⁶⁸ net: ethernet: Add Support for 10G/25G MAC
- [4027f19](#)⁶⁹ net: ethernet: Add 1588 support for 10G/25G MAC
- [f475798](#)⁷⁰ net: ethernet: Fix race condition in the driver for 10G/25G MAC
- [486d636](#)⁷¹ net: ethernet: Add support for 2.5G MAC
- [2857aee](#)⁷² net: ethernet: Fix issues in the driver when DRE is not enabled in the h/w
- [a15cd73](#)⁷³ net: ethernet: Add Clock support
- [9b904af](#)⁷⁴ net: ethernet: Fix Bug in rx reject interrupt handling.

2016.4

- None

2016.3

- Fix kernel crash on the 64-bit platform reported while testing 1588 on Zynq UltraScale+ MPSOC.

Commit Id's:

- [ea9f81c](#)⁷⁵ net: ethernet: xilinx: Fix kernel crash on 64-bit platform

⁶⁶ <https://github.com/Xilinx/linux-xlnx/commit/2c06d773ff537b52f4220b0cdf161724222fa015>
⁶⁷ <https://github.com/Xilinx/linux-xlnx/commit/5b62f654a9600f2c0f686d454b6229588e3c3b23>
⁶⁸ <https://github.com/Xilinx/linux-xlnx/commit/26d9ca49e206263dbe49f2edcad583965453042c>
⁶⁹ <https://github.com/Xilinx/linux-xlnx/commit/819712bd6b00eef8be5a3f13115aa3ee34027f19>
⁷⁰ <https://github.com/Xilinx/linux-xlnx/commit/983a8f96877e5ba8d246231c88319daabf475798>
⁷¹ <https://github.com/Xilinx/linux-xlnx/commit/5e8a3563420238aad936e628bb8431e7f486d636>
⁷² <https://github.com/Xilinx/linux-xlnx/commit/99d053bb0e87207e0177f8ac236477e812857aee>
⁷³ <https://github.com/Xilinx/linux-xlnx/commit/48b19c8e143660322c2b655f6b51104e5a15cd73>
⁷⁴ <https://github.com/Xilinx/linux-xlnx/commit/84fb0cc65aae5970471cbc54b0c89009b9b904af>
⁷⁵ <https://github.com/Xilinx/linux-xlnx/commit/ea9f81ca698a4fb30694bb81a1fd366d8ae58ebb>

11 Related Links

--> <http://www.wiki.xilinx.com/Linux+Drivers>

12 Linux Ethernet Offload Engine driver

Ethernet Offload Engine solution is used with Linux AXI Ethernet driver for XXV Ethernet IP on Zynq Ultrascale+ MPSoC

12.1 Table of Contents

12.2 Introduction

This page gives an overview of Ethernet Offload Engine Linux driver which is available as part of the AMD Linux distribution.

The EOE driver is included in the AXI Ethernet Linux driver and is currently supported with XXV Ethernet subsystem on Zynq Ultrascale+ MPSoC

Paths, files, links and documentation on this page are given relative to the Linux kernel source tree.

12.3 HW IP features

For 10G Ethernet Subsystem (XXV)

- Supported at 10G speed only
- Checksum offload for TX and RX packets
- UDP GSO
- UDP GRO for multiple channels
- UDP packets support ONLY
- Built in Memory option up to 4K Bytes for GRO
- Auto/ User based Selection for GRO port list
- Halt of GRO Merge upon reception of CSUM Error Packets
- Statistic Counters for Dropped, CSUM Error Packets
- Zero Padding removal for IPv4 packets

12.3.1 Features supported in the driver

- Support Checksum Offload
- Support Generic Segmentation Offload for UDP packets
- Support Generic Receive Offload for multiple channels for UDP packets
- Configuring GRO Ports using Ethtool

12.4 Missing Features and Known Issues/Limitations in Driver

- EOE is not support with MCDMA driver in the dmaengine framework at the moment.

12.4.1 Missing Features and Known Issues/Limitations in SDT

- Automatic Devicetree property generation is not available for EOE subsystem. This will be added in a future release.

12.5 Kernel Configuration

The following config options should be enabled in order to build the Axi Ethernet driver

```
CONFIG_ETHERNET
CONFIG_NET_VENDOR_XILINX
CONFIG_XILINX_AXI_EMAC
CONFIG_AXIENET_HAS_MCDMA (Select this option In the design if Axi Ethernet is configured with Axi
MCDMA)
CONFIG_XILINX_AXI_EOE=y
```

[illegible]

12.6 Device-tree

For mor details on EOE bindings, please refer to Documentation/devicetree/bindings/net/xlnx,axi-ethernet.yaml

```
axi_ethernet_eoe: ethernet@80020000 {
    compatible = "xlnx,axi-ethernet-1.00.a";
    interrupts = <0>;
    clock-names = "s_axi_lite_clk", "axis_clk", "ref_clk", "mgt_clk";
    clocks = <&axi_clk>, <&axi_clk>, <&pl_enet_ref_clk>, <&mgt_clk>;
    phy-mode = "mii";
    reg = <0x80020000 0x10000>, <0x80008000 0x8000>;
    reg-names = "mac", "eoe";
    phy-handle = <&phy2>;
    xlnx,rxmem = <0x800>;
}
```

12.6.1 EOE Configured with MCDMA

When Axi EOE is configured with MCDMA, an example devicetree node is as follows: (used with MCDMA driver that is internal to Axi Ethernet driver)

```

ethernet@80020000 {
    axistream-connected = <0x42>;
    clock-frequency = <0x5f5e100>;
    clock-names =
"rx_core_clk\0dclk\0s_axi_aclk\0m_axi_mm2s_aclk\0m_axi_s2mm_aclk\0m_axi_sg_aclk\0s
_axi_lite_aclk";
    clocks = <0x41 0x04 0x48 0x04 0x47 0x40 0x40 0x04 0x47 0x04 0x47>;
    compatible = "xlnx,xxv-ethernet-4.1\0xlnx,xxv-ethernet-1.0";
    device_type = "network";
    local-mac-address = [00 0a 35 00 00 01];
    managed = "in-band-status";
    phy-mode = "10gbase-r";
    reg = <0x00 0x80020000 0x00 0x10000 0x00 0x80008000 0x00 0x8000>;
    reg-names = "mac\0eoe";
    xlnx,rxmem = <0x40000>;
    interrupt-names =
"mm2s_ch10_introut\0mm2s_ch11_introut\0mm2s_ch12_introut\0mm2s_ch13_introut\0mm2s_
ch14_introut\0mm2s_ch15_introut\0mm2s_ch16_introut\0mm2s_ch1_introut\0mm2s_ch2_int
rout\0mm2s_ch3_introut\0mm2s_ch4_introut\0mm2s_ch5_introut\0mm2s_ch6_introut\0mm2s
_ch7_introut\0mm2s_ch8_introut\0mm2s_ch9_introut\0s2mm_ch10_introut\0s2mm_ch11_int
rout\0s2mm_ch12_introut\0s2mm_ch13_introut\0s2mm_ch14_introut\0s2mm_ch15_introut\0
s2mm_ch16_introut\0s2mm_ch1_introut\0s2mm_ch2_introut\0s2mm_ch3_introut\0s2mm_ch4_
introut\0s2mm_ch5_introut\0s2mm_ch6_introut\0s2mm_ch7_introut\0s2mm_ch8_introut\0s
2mm_ch9_introut";
    interrupt-parent = <0x05>;
    interrupts = <0x00 0x59 0x04 0x00 0x59 0x04 0x00 0x59 0x04 0x00 0x59
0x04 0x00 0x59 0x04 0x00 0x59 0x04 0x00 0x59 0x04 0x00 0x59 0x04 0x00 0x59 0x04
0x00 0x59 0x04 0x00 0x59 0x04 0x00 0x59 0x04 0x00 0x59 0x04 0x00 0x59 0x04 0x00
0x59 0x04 0x00 0x59 0x04 0x00 0x5a 0x04 0x00 0x5a 0x04 0x00 0x5a 0x04 0x00 0x5a
0x04 0x00 0x5a 0x04 0x00 0x5a 0x04 0x00 0x5a 0x04 0x00 0x5a 0x04 0x00 0x5a 0x04
0x00 0x5a 0x04 0x00 0x5a 0x04 0x00 0x5a 0x04 0x00 0x5a 0x04 0x00 0x5a 0x04 0x00
0x5a 0x04 0x00 0x5a 0x04>;
    xlnx,channel-ids = "1\02\03\04\05\06\07\08\09\0a\0b\0c\0d\0e\0f\010";
    xlnx,include-dre;
    xlnx,has-hw-offload;
    xlnx,rx-hw-offload = <0x02>;
    xlnx,tx-hw-offload = <0x02>;
    phandle = <0xab>;

    mdio {
        #address-cells = <0x01>;
        #size-cells = <0x00>;
        phandle = <0xac>;
    };
};

```

12.7 Performance

These benchmark performance numbers were obtained by connecting AMD boards to thirdparty NIC cards over Linux PCs/server machines (Ubuntu/Red Hat Enterprise).

The tools used are aperf (Refer to tool information below).

Performance benchmark numbers mentioned in below tables are for reference and dependent on multiple factors i.e setup , vivado design configuration etc.

NOTE: CPU utilization reported in below performance tables is an aggregate of all CPU's. i.e on ZynqMP platform, it reports combined utilization of all four A53 cores.

12.7.1 10G Ethernet with AXIMCDMA

Kernel version: 6.6

AXI 10G Ethernet Subsystem : TX and RX checksum offload enabled.

ZynqMP

Board: ZCU102 board (production silicon) + SFP Module

For 1 GRO Port

	UDP (Gbps)			
MTU	TX	CPU(%)	RX	CPU(%)
1500	9.56	20.82	9.56	31.03

NOTE- EOE Supports 15 Ports for GRO.

Setup Details

Host setup: SuperMicro (H12SSL-NT)

aperf: iperf 3.9 (cJSON 1.7.13)

OS : Ubuntu 20.04.1 LTS (Linux kernel 5.15.0-107-generic)

NIC (Solarflare XtremeScale X2522 10/25GbE Ultra-Low Latency Network Adapter) : Default

12.7.2 Performance Test

How to Run

Server:

aperf -s -p <Port Number>

Client:

aperf -p <Port Number> -c <SERVER_IP> -b <bandwidth> -u -l <Buffer Length> -G <Packet length>

12.8 Mainline status

The EOE driver is currently NOT in mainline

12.9 Change Log

2024.1

- First release adds Support for Ethernet Offload Engine. Generic Segmentation Offload, and multi-port Generic Receive Offload

Commits

https://github.com/Xilinx/linux-xlnx/commits/xilinx-v2024.2/drivers/net/ethernet/xilinx/xilinx_axienet_eoe.c

12.10 Related Links

- [Linux AXI Ethernet driver](#)
- [Linux Drivers](#)⁷⁶

⁷⁶ <https://xilinx-wiki.atlassian.net/wiki/spaces/A/pages/18841873/Linux+Drivers>